

Python Mini Project (EDA Based)

Case Study: NYC Taxi Trip Duration Analysis

Background

New York City generates millions of taxi trips every month. Each trip produces valuable data related to time, location, passengers, and travel duration. Urban planners and transport authorities analyze such data to understand traffic patterns, congestion behavior, and commuter habits.

In this mini project, you will work with a real-world dataset used in a Kaggle competition. Your task is to perform Exploratory Data Analysis (EDA) to uncover meaningful patterns and insights from the data.

Learning Objectives

- Apply Python libraries such as Pandas, NumPy, Matplotlib, and Seaborn for data analysis
- Understand and clean real-world, large-scale datasets
- Perform univariate, bivariate, and multivariate analysis
- Generate insights using visual and statistical exploration
- Communicate findings clearly using structured analysis

Dataset Information

Dataset Name: NYC Taxi Trip Duration (Kaggle Competition Dataset)

The dataset contains taxi trip records with information such as pickup time, drop-off time, passenger count, pickup and drop-off coordinates, and trip duration.

Project Instructions

Phase 1: Data Understanding

- Load the dataset and inspect its structure
- Display the first and last few records
- Check dataset dimensions and column data types

Phase 2: Data Quality Checks

- Identify missing values and inconsistencies
- Detect invalid or unrealistic entries
- Comment on data quality issues found

Phase 3: Data Cleaning

- Handle missing or invalid values appropriately
- Remove or treat outliers where necessary
- Justify all cleaning decisions in markdown

Phase 4: Feature Engineering for EDA

- Extract useful time-based features from datetime columns
- Create additional analytical features if required

Phase 5: Univariate Analysis

- Analyze distributions of important variables
- Use appropriate plots to support observations

Phase 6: Bivariate Analysis

- Analyze relationships between two variables
- Use plots and summary statistics to support findings

Phase 7: Multivariate Analysis

- Explore interactions among multiple variables
- Use grouping, pivot tables, or heatmaps where appropriate

Phase 8: Insight Generation

- Write at least 10 clear and data-backed insights
- Each insight must be supported by analysis or visualization

Phase 9: Final Conclusion

- Summarize key findings
- Discuss how these insights could help urban transport planning
- Mention any limitations of the dataset

Submission Guidelines

- Submit a Jupyter Notebook (.ipynb) file
- Use proper headings and markdown explanations
- All plots must have titles, labels, and legends where applicable
- Do not include machine learning models

Evaluation Criteria

- Data cleaning and preparation
- Depth and correctness of EDA
- Quality of visualizations
- Clarity of insights and conclusions
- Overall structure and presentation